

Machine Learning Meets Riyadh Real Estate: An Empirical Perspective

DONE BY

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INTRODUCTION

Our project aims to understand the real estate market in Saudi Arabia, specifically in Riyadh, by examining price differences between residential and commercial properties, as well as residential unit prices across different areas of the city. We collect detailed data on rental prices in Riyadh, categorized by region, and analyze the factors driving price changes and trends in the Saudi real estate market. This analysis provides investors and renters with a clear view of price trends and market variations based on location and property type, supporting informed investment decisions.

OBJECTIVE

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- Understand Market Trends: Investigate factors that drive price changes in Riyadh's real estate market, offering insights into current trends and future price projections.
- Support Informed Decision-Making: Provide investors and renters with data-driven insights that help them make well-informed choices based on property type and location trends in the Riyadh market.

RELATED WORK

- Khaled et al. (2020) explore how Saudi Vision 2030 influences real estate demand and pricing in Riyadh.
- Malik et al. (2018) analyze the effect of housing features like bedrooms, pools, and private roofs on property prices.

DATA

The data we collected is sourced from **Aqar** platform, which provides reliable information about real estate prices across Saudi Arabia.

Title	Description
Title	Akar2024
Purpose	For an empirical study of Riyadh real estate
Owner	عقاري. ai
Target	price
Restriction	Only for members of عقاري. ai
Date of collection	2024/10/24
Scraped from	/https://sa.aqar.fm
Number of attributes	37
Number of records	111297
Category	Real estate
Location	Riyadh, Saudi Arabia

Figure 1 Metadata

METHODOLOGY

- Data Extraction & Processing:** Extracted raw data from the Aqar platform, followed by intensive data cleaning, filtering, and outlier removal to ensure data quality and reliability.
- Exploratory Data Analysis (EDA):** Conducted visualizations to identify trends and insights, helping to answer specific research questions and facilitate comparisons within the dataset.
- Modeling & Evaluation:** Used Gradient Boosting Regression with hyperparameter tuning to predict property prices. Quantile regression models were added to provide price ranges, ensuring accurate and reliable predictions.
- System Development:** Developed a predictive system using Hugging Face and Gradio, allowing user input-based predictions. Coding of the entire project was done primarily in R, with supplemental Python.

HYPOTHESES

- Focusing on The Area, number of rooms and their kind, and the location is sufficient to make a predictive model that predict the price range of the property.

We used a **Gradient Boosting Regressor**, **Random Forest Regressor**, and **Linear Regressor** to predict property prices based on features such as the number of living rooms, kitchens, rooms, bathrooms (WC), area, category, and location. The dataset is split into training and testing sets, and hyperparameter tuning is performed using RandomizedSearchCV to optimize the models. Additionally, quantile regression models are trained to predict the lower and upper bounds of the price range.

RESULTS

{f"Accuracy within predicted range: {range_accuracy:.2f}%"}

Accuracy within predicted range: 79.21%

Figure 2 Prediction Accuracy Validated

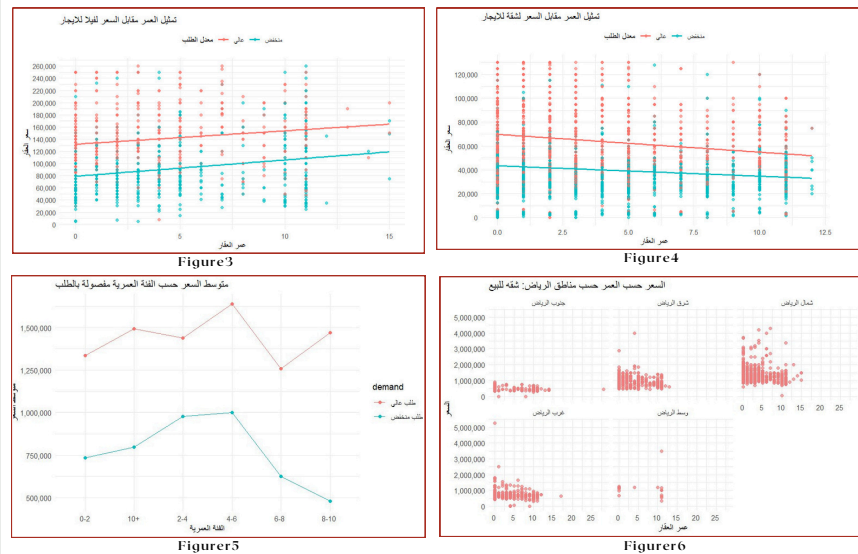
In Figure 2, we present the results of using a **Gradient Boosting Regressor** to validate the hypothesis that area, number of rooms, room types, and location are sufficient predictors of property prices. The model achieved an accuracy of 79.2%, highlighting its effectiveness in capturing key determinants of pricing trends.

- As properties age, the influence of age on price is moderated by other factors, such as city and location. Older properties in high-demand locations tend to retain their value better than those in less desirable areas.

Rented Properties in Riyadh: The hypothesis is partially confirmed. Age is a weak predictor of price, primarily due to Riyadh's high demand for rental properties. In contrast, in Dammam, the negative effect of age on price is twice as significant.

Buying Properties in Riyadh: The hypothesis is confirmed. Older properties in high-demand areas retain their value, highlighting the crucial role of location in preserving property value over time.

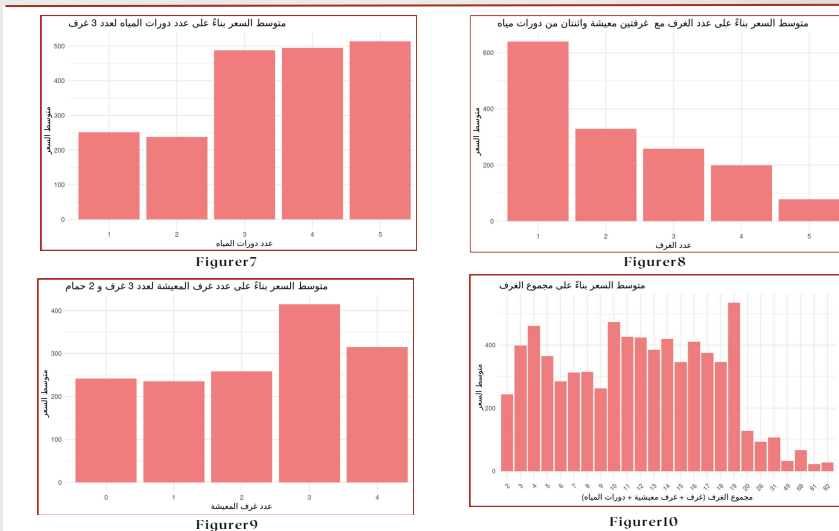
RESULTS



- the number and type of rooms in a property (e.g., bedrooms, living rooms, etc.) significantly affect its price, but that this effect decreases beyond a certain point.

Property prices rise with more living rooms and bathrooms but decrease with an excess of general rooms. Total room count initially increases prices, but properties with 20+ rooms see a sharp drop, indicating a preference for balanced layouts over excessive room counts. This trend suggests that buyers may prefer properties with balanced features rather than an excessive number of general rooms, which might indicate inefficiency in the property layout.

RESULTS



CONCLUSION

- The model achieved 79.21% accuracy, supporting the hypothesis that area, room details, and location are sufficient to predict property prices effectively.
- The hypothesis that location, area, and room types significantly impact property prices is confirmed, while the effect of property age varies by city, being less significant in high-demand areas like Riyadh.
- The hypothesis Property prices increase with more living rooms and bathrooms but decrease as the number of general rooms rises. Combining all rooms, prices initially increase but drop sharply for properties with 20 or more rooms.

FUTURE WORK

- Collect additional data to expand the analysis beyond Riyadh, covering all major cities in Saudi Arabia.
- Improve the accuracy and reliability of our Real Estate Price Prediction system.
- Develop a more advanced system that delivers personalized insights for investors, renters, and market analysts, focusing on different cities.

REFERENCE

- Khaled, A., Alshahrani, T. and Alqahtani, M. (2020) 'Economic development and real estate markets under Saudi Vision 2030: A focus on Riyadh,' Middle East Development Journal, 12(2), pp. 93–112.
- Malik, R., Khan, M. and Ahmed, S. (2018) 'Impact of housing features on property pricing in urban settings: A data-driven approach,' Journal of Urban Economics, 92(3), pp. 303–318.

